



$y_1, \dots, y_n$

$y_i \in \{H, T\}$

$H = 1$
 $T = 0$

$y = \# \text{ Heads}$
 $n = \# \text{ Flips}$

$\Theta = [0, 1]$ parameter space

$$\mathcal{F} := \left\{ \binom{n}{y} \theta^y (1-\theta)^{n-y} \mid \theta \in \Theta \right\}$$

$$\binom{n}{y} \left(\frac{1}{2}\right)^y \left(\frac{1}{2}\right)^{n-y}$$

$$\binom{n}{y} \left(\frac{1}{3}\right)^y \left(\frac{2}{3}\right)^{n-y}$$

Case 1: $f(y) \in \mathcal{F}$ $\exists \theta_0 \in \Theta$ s.t. $f(y) = \binom{n}{y} \theta_0^y (1-\theta_0)^{n-y}$

Case 2: Misspecification of the model

$$f(y) \notin \mathcal{F}$$

but then there is some function $p(\theta_0 | y)$ that's the closest to $f(y)$.

Sketch Proof 1:

Assumptions:

1. Θ is finite $\Theta = \{\theta_0, \theta_1, \dots, \theta_k\}$
2. θ_0 is the KL(θ) minimizer
3. $p(\theta = \theta_0) > 0$ *it's within our prior*

not necessarily the smallest

Aim: show $p(\theta = \theta_0 | y) \rightarrow 1$ as $n \rightarrow \infty$

Let $\theta \neq \theta_0$

$$\begin{aligned} \log \left(\frac{p(\theta | y)}{p(\theta_0 | y)} \right) &= \log \left(\frac{p(\theta) \prod_{i=1}^n p(y_i | \theta)}{p(\theta_0) \prod_{i=1}^n p(y_i | \theta_0)} \right) \\ &= \log \left(\frac{p(\theta)}{p(\theta_0)} \right) + \sum_{i=1}^n \log \left(\frac{p(y_i | \theta)}{p(y_i | \theta_0)} \right) \end{aligned}$$

Observe:

$$\begin{aligned} \sum_{i=1}^n \log \left(\frac{p(y_i | \theta)}{p(y_i | \theta_0)} \right) &= \log(p(y_i | \theta)) - \log(p(y_i | \theta_0)) \\ &= \log(p(y_i | \theta)) - \log(f(y_i)) + \log(f(y_i)) - \log(p(y_i | \theta_0)) \\ &= \log \left(\frac{f(y_i)}{p(y_i | \theta_0)} \right) - \log \left(\frac{f(y_i)}{p(y_i | \theta)} \right) \end{aligned}$$

Recall y_i are i.i.d. So by the law of large numbers

$$\begin{aligned} \log \left(\frac{p(\theta|y)}{p(\theta_0|y)} \right) &= \log \left(\frac{p(\theta)}{p(\theta_0)} \right) + \sum_{i=1}^n \left[\log \left(\frac{f(y_i)}{p(y_i|\theta_0)} \right) - \log \left(\frac{f(y_i)}{p(y_i|\theta)} \right) \right] \\ &= \log \left(\frac{p(\theta)}{p(\theta_0)} \right) + n E \left(\log \left(\frac{f(y_i)}{p(y_i|\theta_0)} \right) \right) - n E \left(\log \left(\frac{f(y_i)}{p(y_i|\theta)} \right) \right) \\ &= \underbrace{\log \left(\frac{p(\theta)}{p(\theta_0)} \right)}_{\text{constant}} + n \left[\underbrace{KL(\theta_0)}_{\text{minimizer}} - KL(\theta) \right] \end{aligned}$$

Since $\theta \neq \theta_0$ and θ_0 is a minimizer of $KL(\cdot)$

$$\log \left(\frac{p(\theta|y)}{p(\theta_0|y)} \right) \rightarrow -\infty \quad \text{as } n \rightarrow \infty$$

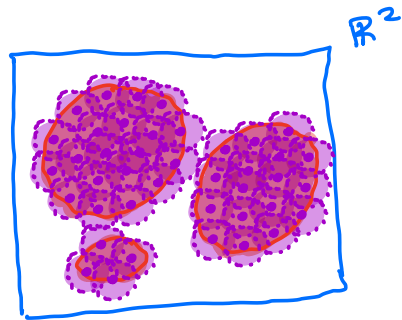
$$\Rightarrow \frac{p(\theta|y)}{p(\theta_0|y)} \rightarrow 0 \quad \Rightarrow p(\theta|y) \rightarrow 0$$

Because probabilities sum to 1, $\boxed{p(\theta_0|y) \rightarrow 1}$.

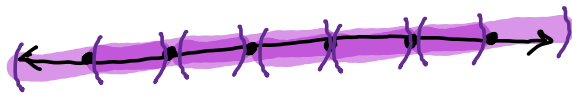
Hence, the posterior distribution converges.

A set (H) is compact if every open cover contains a finite subcover.

Since $(H) \subseteq \mathbb{R}^n$, we picture this as closed and bounded.



Counter $\mathbb{Z}$ in $\mathbb{R}$



$$A = \{ (z-1, z+1) \mid z \in \mathbb{Z} \}$$

I cannot delete any set in this open cover because only one covers each point.

Sketch proof for Theorem 2

Let (H) be a compact set and define $\mathcal{A}$ to be an open cover such that only one set $A_0 \in \mathcal{A}$ contains θ_0 .

Since (H) is compact, there exists a finite subcover.

$\{A_0, A_1, \dots, A_k\}$ where A_0 is the set previously specified.

Use theorem 1's argument.

Let $A \neq A_0$

$$\log \left(\frac{p(\theta \in A | y)}{p(\theta \in A_0 | y)} \right) \approx \log \left(\frac{p(\theta \in A)}{p(\theta \in A_0)} \right) + n \mathbb{E} \left(\frac{p(y; \theta \in A)}{p(y; \theta \in A_0)} \right)$$

$$p(\theta \in A | y) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

$$p(\theta \in A_0 | y) \rightarrow 1 .$$